AN ETALE-SMOOTH LOCAL PROPERTY ON THE SOURCE AND TARGET IS CHECKED ON ONE PRESENTATION OF A DM MORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The notion of a property of morphisms of algebraic spaces being etale-smooth local on the source-and-target, in the three forms recorded in the chapter on descent on spaces. Invoked inside the counted proof (source lines 7851-7860 and 7876-7879) as cross-chapter pointers to the definition and to two lemmas. Allows the property to be moved along the smooth base change and along the etale covering used in the comparison.
- The description of a DM morphism through the existence of an etale presentation of the fibre product over a smooth presentation of the target. Invoked at the start of the counted proof (source lines 7798-7800) as a pointer to the author lemma lemma-DM (labelled separately in the same chapter). Produces the diagram of the second condition out of the DM hypothesis.

Source reference. Original lemma lemma-DM: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 7800–7800. Navigation pointer only: the counted text refers to the Stacks item lemma-DM, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference spaces-descent-definition-etale-smooth-local-source-target: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 7759–7759. Navigation pointer only: the counted text refers to the Stacks item spaces-descent-definition-etale-smooth-local-source-target, which belongs to another chapter of the project and is not part of this card. The pointer

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Source reference. Original reference spaces-descent-lemma-etale-smooth-local-source-target-characterize: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 7885–7885. Navigation pointer only: the counted text refers to the Stacks item spaces-descent-lemma-etale-smooth-local-source-target-characterize, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference spaces-descent-lemma-etale-smooth-local-source-target-implies: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 7853–7853. Navigation pointer only: the counted text refers to the Stacks item spaces-descent-lemma-etale-smooth-local-source-target-implies, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference stacks-properties-section-properties-morphisms: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 248–248. Navigation pointer only: the counted text refers to the Stacks item stacks-properties-section-properties-morphisms, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $\mathcal{P}$ be a property of morphisms of algebraic spaces which is étale-smooth local on the source-and-target. Let $f : \mathcal{X} \to \mathcal{Y}$ be a DM morphism of algebraic stacks. Consider commutative diagrams*

$$\begin{array}{ccc} U & \xrightarrow[h]{} & V \\ {\scriptstyle a}\downarrow & & \downarrow{\scriptstyle b} \\ \mathcal{X} & \xrightarrow{f} & \mathcal{Y} \end{array}$$

where U and V are algebraic spaces, $V \to \mathcal{Y}$ is smooth, and $U \to \mathcal{X} \times_{\mathcal{Y}} V$ is étale. The following are equivalent

1. *for any diagram as above the morphism h has property $\mathcal{P}$, and*
2. *for some diagram as above with $a : U \to \mathcal{X}$ surjective the morphism h has property $\mathcal{P}$.*

If $\mathcal{X}$ and $\mathcal{Y}$ are representable by algebraic spaces, then this is also equivalent to f (as a morphism of algebraic spaces) having property $\mathcal{P}$. If $\mathcal{P}$ is also preserved under any base change, and fppf local on the base, then for morphisms f which are representable by algebraic spaces this is also equivalent to f having property $\mathcal{P}$ in the sense of Properties of Stacks, Section stacks-properties-section-properties-morphisms.

Proof. Let us prove the implication (1) $\Rightarrow$ (2). Pick an algebraic space V and a surjective and smooth morphism $V \to \mathcal{Y}$. As f is DM there exists a scheme U and a surjective étale morphism $U \to V \times_{\mathcal{Y}} \mathcal{X}$, see Lemma lemma-DM. Thus we see that (2) holds. Note that $U \to \mathcal{X}$ is surjective and smooth as well, as a composition of the base change $\mathcal{X} \times_{\mathcal{Y}} V \to \mathcal{X}$ and the chosen map $U \to \mathcal{X} \times_{\mathcal{Y}} V$. Hence we obtain a diagram as in (1). Thus if (1) holds, then $h : U \to V$ has property $\mathcal{P}$, which means that (2) holds as $U \to \mathcal{X}$ is surjective.

Conversely, assume (2) holds and let U, V, a, b, h be as in (2). Next, let U', V', a', b', h' be any diagram as in (1). Picture

$$\begin{array}{ccc} U & \xrightarrow[h]{} & V \\ \downarrow & & \downarrow \\ \mathcal{X} & \xrightarrow{f} & \mathcal{Y} \end{array} \qquad \begin{array}{ccc} U' & \xrightarrow[h']{} & V' \\ \downarrow & & \downarrow \\ \mathcal{X} & \xrightarrow{f} & \mathcal{Y} \end{array}$$

To show that (2) implies (1) we have to prove that h' has $\mathcal{P}$. To do this consider the commutative diagram

$$\begin{array}{ccccc} U & \longleftarrow & U \times_{\mathcal{X}} U' & \longrightarrow & U' \\ \downarrow{\scriptstyle h} & & \downarrow{\scriptstyle (h,h')} & & \downarrow{\scriptstyle h'} \\ V & \longleftarrow & V \times_{\mathcal{Y}} V' & \longrightarrow & V' \end{array}$$

of algebraic spaces. Note that the horizontal arrows are smooth as base changes of the smooth morphisms $V \to \mathcal{Y}$, $V' \to \mathcal{Y}$, $U \to \mathcal{X}$, and $U' \to \mathcal{X}$. Note that the squares

$$\begin{array}{ccc} U & \longleftarrow & U \times_{\mathcal{X}} U' \\ \downarrow & & \downarrow \\ V \times_{\mathcal{Y}} \mathcal{X} & \longleftarrow & V \times_{\mathcal{Y}} U' \end{array} \qquad \begin{array}{ccc} U \times_{\mathcal{X}} U' & \longrightarrow & U' \\ \downarrow & & \downarrow \\ U \times_{\mathcal{Y}} V' & \longrightarrow & \mathcal{X} \times_{\mathcal{Y}} V' \end{array}$$

are cartesian, hence the vertical arrows are étale by our assumptions on U', V', a', b', h' and U, V, a, b, h. Since $\mathcal{P}$ is smooth local on the target by Descent on Spaces, Lemma spaces-descent-lemma-etale-smooth-local-source-target-implies part (2) we see that the base change $t : U \times_{\mathcal{Y}} V' \to V \times_{\mathcal{Y}} V'$ of h has $\mathcal{P}$. Since $\mathcal{P}$ is étale local on the source by Descent on Spaces, Lemma spaces-descent-lemma-etale-smooth-local-source-target-implies part (1) and $s : U \times_{\mathcal{X}} U' \to U \times_{\mathcal{Y}} V'$ is étale, we see the morphism $(h, h') = t \circ s$ has $\mathcal{P}$. Consider the diagram

$$\begin{array}{ccc} U \times_{\mathcal{X}} U' & \xrightarrow[(h,h')]{} & V \times_{\mathcal{Y}} V' \\ \downarrow & & \downarrow \\ U' & \xrightarrow{h'} & V' \end{array}$$

The left vertical arrow is surjective, the right vertical arrow is smooth, and the induced morphism

$$U \times_{\mathcal{X}} U' \longrightarrow U' \times_{V'} (V \times_{\mathcal{Y}} V') = V \times_{\mathcal{Y}} U'$$

is étale as seen above. Hence by Descent on Spaces, Definition spaces-descent-definition-etale-smooth-local-source-target part (3) we conclude that h' has $\mathcal{P}$. This finishes the proof of the equivalence of (1) and (2).

If $\mathcal{X}$ and $\mathcal{Y}$ are representable, then Descent on Spaces, Lemma spaces-descent-lemma-etale-smooth-local-source-target-characterize applies which shows that (1) and (2) are equivalent to f having $\mathcal{P}$.

Finally, suppose f is representable, and U, V, a, b, h are as in part (2) of the lemma, and that $\mathcal{P}$ is preserved under arbitrary base change. We have to show that for any scheme Z and morphism $Z \to \mathcal{X}$ the base change $Z \times_{\mathcal{Y}} \mathcal{X} \to Z$ has property

$\mathcal{P}$. Consider the diagram

$$\begin{array}{ccc} Z \times_{\mathcal{Y}} U & \longrightarrow & Z \times_{\mathcal{Y}} V \\ \downarrow & & \downarrow \\ Z \times_{\mathcal{Y}} \mathcal{X} & \longrightarrow & Z \end{array}$$

Note that the top horizontal arrow is a base change of h and hence has property $\mathcal{P}$. The left vertical arrow is surjective, the induced morphism

$$Z \times_{\mathcal{Y}} U \longrightarrow (Z \times_{\mathcal{Y}} \mathcal{X}) \times_Z (Z \times_{\mathcal{Y}} V)$$

is étale, and the right vertical arrow is smooth. Thus Descent on Spaces, Lemma spaces-descent-lemma-etale-smooth-local-source-target-characterize kicks in and shows that $Z \times_{\mathcal{Y}} \mathcal{X} \to Z$ has property $\mathcal{P}$. $\square$